

Polynomial Regression for Renewable Geothermal Systems: High-Dimensional Surface Optimization and Deep Reservoir Mapping

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GitHub Repository: https://github.com/Jdsb06/ML_assignment

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Abstract

In this work, we develop and evaluate regularized polynomial regression architectures to solve two distinct real-world geothermal energy engineering problems: (1) multi-stage steam turbine optimization (var1) governed by 6 operational parameters, and (2) subterranean thermal anomaly mapping (var2) governed by 3D spatial coordinates. We formulate the regression tasks under the lens of the bias-variance tradeoff and address the severe ill-conditioning and parameter proliferation associated with high-degree multivariate monomial expansions. For Phase 1, where unconstrained ordinary least squares (OLS) encounters catastrophic overfitting due to 461 parameters on $N = 1000$ points and a notable boundary clamping shift (51.25% of test coordinates clamped at ± 1 versus 31.17% in train), we demonstrate that a **Degree 5 Polynomial with Feature Standardization and L_1 Lasso Regularization** ($\alpha = 0.015$) achieves optimal generalization, obtaining a 10-fold cross-validation Mean Squared Error (MSE) of 0.3314 ± 0.0424 and an R^2 score of 0.9668 ± 0.0055 , effectively pruning 79.0% of redundant interactions (retaining 97 active terms). For Phase 2, representing complex geological subsurface contours, we demonstrate that a **Degree 10 Polynomial with L_2 Ridge Regularization** ($\alpha = 0.7$) achieves a 10-fold cross-validation MSE of 0.2652 ± 0.0729 and R^2 of 0.9951 ± 0.0014 , closely rivaled by a parsimonious Degree 8 Ridge model (MSE = 0.2691 ± 0.0656 , well within fold noise). All code, experimental research scripts, trained pipelines, evaluation benchmarks, and predictions are publicly available at the project repository.

1. Introduction & Problem Formulation

Geothermal power generation involves intricate interactions between thermodynamic surface facilities and heterogeneous subterranean reservoirs. Modeling these systems accurately is critical for maximizing megawatt electrical yield and minimizing expensive drilling operations. In this investigation, we explore polynomial regression models on two personalized calibration datasets for roll number IMT2024022.

1.1. Phase 1: Steam Turbine Optimization (var1)

A surface geothermal plant operates a multi-stage steam turbine whose thermodynamic efficiency dictates the Net Power Score ($y \in \mathbb{R}$). The turbine operating state is captured by

$D = 6$ percentage deviation parameters:

- x_1 : High-pressure steam valve adjustment
- x_2 : Condenser coolant flow rate adjustment
- x_3 : Re-injection pump hydraulic pressure
- x_4 : Turbine blade pitch angle
- x_5 : Non-condensable gas exhaust valve rate
- x_6 : Steam inlet pressure adjustment

Historical calibration logs provide $N_{\text{train}} = 1000$ records, and the objective is to predict y across $N_{\text{test}} = 1000$ test settings using moderate-degree polynomial regression (up to degree 10).

1.2. Phase 2: Subterranean Thermal Mapping (var2)

Phase 2 focuses on identifying optimal drilling coordinates for new production wells within an exploration sector. A sensor network records the Thermal Anomaly Score (y) across 3D spatial offsets from a central landmark:

- x_1 : East-West coordinate offset (meters)
- x_2 : North-South coordinate offset (meters)
- x_3 : Vertical depth offset relative to basecamp (meters)

Core samples provide $N_{\text{train}} = 1000$ sparse spatial measurements, and the goal is to predict y over $N_{\text{test}} = 1000$ planned drilling coordinates using polynomials up to degree 20.

2. Mathematical Framework & The Curse of Dimensionality

2.1. Multivariate Polynomial Feature Mapping

Given an input vector $\mathbf{x} = [x_1, \dots, x_D]^T \in \mathbb{R}^D$, a polynomial hypothesis of degree d maps $\mathbf{x}$ into a high-dimensional feature space via the monomial basis expansion $\phi_d(\mathbf{x})$:

$$\phi_d(\mathbf{x}) = \left[\prod_{j=1}^D x_j^{k_j} : \sum_{j=1}^D k_j \leq d, k_j \in \mathbb{N}_0 \right]^T \quad (1)$$

The regression hypothesis is linear in the expanded basis:

$$\hat{y}(\mathbf{x}) = \mathbf{w}^T \phi_d(\mathbf{x}) + b \quad (2)$$

The total number of parameters (excluding the intercept b) is given by the combinatoric formula:

$$p(D, d) = \binom{D+d}{d} - 1 \quad (3)$$

2.2. Curse of Dimensionality in Multivariate Systems

The scaling behavior differs radically between Phase 1 ($D = 6$) and Phase 2 ($D = 3$):

- For $D = 6$: Degree 1 yields $p = 6$; degree 2 yields $p = 27$; degree 3 yields $p = 83$; degree 4 yields $p = 209$; degree 5 yields $p = 461$; degree 6 yields $p = 923$; and degree 10 yields $p = 8007$.
- For $D = 3$: Degree 2 yields $p = 9$; degree 4 yields $p = 34$; degree 6 yields $p = 83$; degree 8 yields $p = 164$; degree 10 yields $p = 285$; degree 12 yields $p = 454$; and degree 20 yields $p = 1770$.

In Phase 1, setting $d \geq 6$ causes the number of parameters p to approach or exceed the number of training samples ($N_{\text{train}} = 1000$, and in cross-validation $N_{\text{fold}} = 900$). Under Ordinary Least Squares (OLS):

$$\hat{\mathbf{w}}_{\text{OLS}} = (\Phi^T \Phi)^{-1} \Phi^T \mathbf{y} \quad (4)$$

When $p \approx N$, the Gram matrix $\Phi^T \Phi$ becomes severely ill-conditioned (condition number $\kappa(\Phi^T \Phi) \rightarrow \infty$). Consequently, the parameter variance explodes:

$$\text{Var}(\hat{\mathbf{w}}_{\text{OLS}}) = \sigma^2 (\Phi^T \Phi)^{-1} \quad (5)$$

leading to catastrophic out-of-sample prediction error.

2.3. Standardization and Regularization Strategies

Because input variables $x_j \in [-1, 1]$, computing higher powers x_j^k induces vastly disparate numerical scales (e.g., $x_j^{10} \ll x_j$). Therefore, standardizing all polynomial basis columns to zero mean and unit variance is strongly advisable prior to applying penalized regression:

$$\tilde{\phi}_{d,j}(\mathbf{x}) = \frac{\phi_{d,j}(\mathbf{x}) - \mu_j}{\sigma_j} \quad (6)$$

To control variance and prevent overfitting, we explore two continuous regularization paradigms:

1. Ridge Regression (L_2 Penalty):

$$\mathcal{L}_{\text{Ridge}}(\mathbf{w}) = \|\mathbf{y} - \tilde{\Phi} \mathbf{w}\|_2^2 + \lambda \|\mathbf{w}\|_2^2 \quad (7)$$

$$\hat{\mathbf{w}}_{\text{Ridge}} = (\tilde{\Phi}^T \tilde{\Phi} + \lambda \mathbf{I})^{-1} \tilde{\Phi}^T \mathbf{y} \quad (8)$$

Ridge adds a strictly positive perturbation $\lambda \mathbf{I}$ to the diagonal of the Gram matrix, guaranteeing invertibility and bounding the condition number regardless of collinearity.

2. Lasso Regression (L_1 Penalty):

$$\mathcal{L}_{\text{Lasso}}(\mathbf{w}) = \frac{1}{2N} \|\mathbf{y} - \tilde{\Phi} \mathbf{w}\|_2^2 + \alpha \|\mathbf{w}\|_1 \quad (9)$$

Due to the non-differentiable singularity of the L_1 ball at coordinate axes, Lasso performs automatic feature selection by shrinking irrelevant interaction terms exactly to zero.

3. Exploratory Data Analysis

A rigorous preliminary audit was conducted on both datasets corresponding to roll number IMT2024022:

- **Data Integrity:** Neither dataset contains missing, null, or infinite values. Both training sets contain exactly 1000 observations; both test sets contain 1000 observations.
- **Phase 1 Domain:** Feature variables x_1 through x_6 are bounded within $[-1.000, 1.000]$ with empirical means close to zero ($\mu_{x_j} \in [-0.044, -0.001]$) and standard deviations ≈ 0.71 . The target y has mean $\mu = 0.8966$, standard deviation $\sigma = 3.1867$, spanning $[-8.235, 13.248]$.
- **Phase 2 Domain:** Coordinate offsets x_1, x_2, x_3 are bounded within $[-1.000, 1.000]$ with means ≈ 0.00 and standard deviations ≈ 0.68 . The target y exhibits higher dynamic range: mean $\mu = 2.3426$, standard deviation $\sigma = 7.4816$, spanning $[-29.710, 39.179]$.
- **Boundary Clamping in Both Phases & Phase 1 Covariate Shift:** A systematic coordinate audit reveals that boundary clamping at ± 1.0 is present in *both* datasets:
 - In **Phase 2**, coordinate clamping occurs symmetrically at domain boundaries: 26.17% of coordinates (785/3000) in train and 24.83% (745/3000) in test are clamped at ± 1.0 . The observation-level distribution across clamping strata $k \in \{0, 1, 2, 3\}$ is remarkably balanced (40.5% vs 42.9% at $k = 0$, 42.1% vs 41.5% at $k = 1$, 15.8% vs 13.8% at $k = 2$, and 1.6% vs 1.8% at $k = 3$), indicating absence of significant covariate shift across partitions.
 - In **Phase 1**, however, an acute covariate shift exists: 31.17% of training coordinates (1870/6000) are clamped at ± 1.0 , whereas in the test set, clamping surges to 51.25% (3075/6000). At the observation level, only 28.7% of training points exhibit ≥ 3 clamped coordinates, whereas 66.5% of test points reside in these high-boundary strata. Because multivariate monomials evaluate to extreme amplitudes at ± 1.0 , unregularized models face severe boundary extrapolation risk.

4. Experimental Methodology

To rigorously guard against data leakage and hyperparameter overfitting, all modeling steps were integrated into composite Scikit-Learn Pipeline objects consisting of:

1. Polynomial expansion of degree d without raw intercept.
2. Feature standardization (`StandardScaler`) fitted strictly on training folds.
3. Estimator (`LinearRegression`, `Ridge`, or `Lasso`).

Validation was performed using 10-Fold Cross-Validation with shuffle enabled and a fixed pseudo-random seed (42).

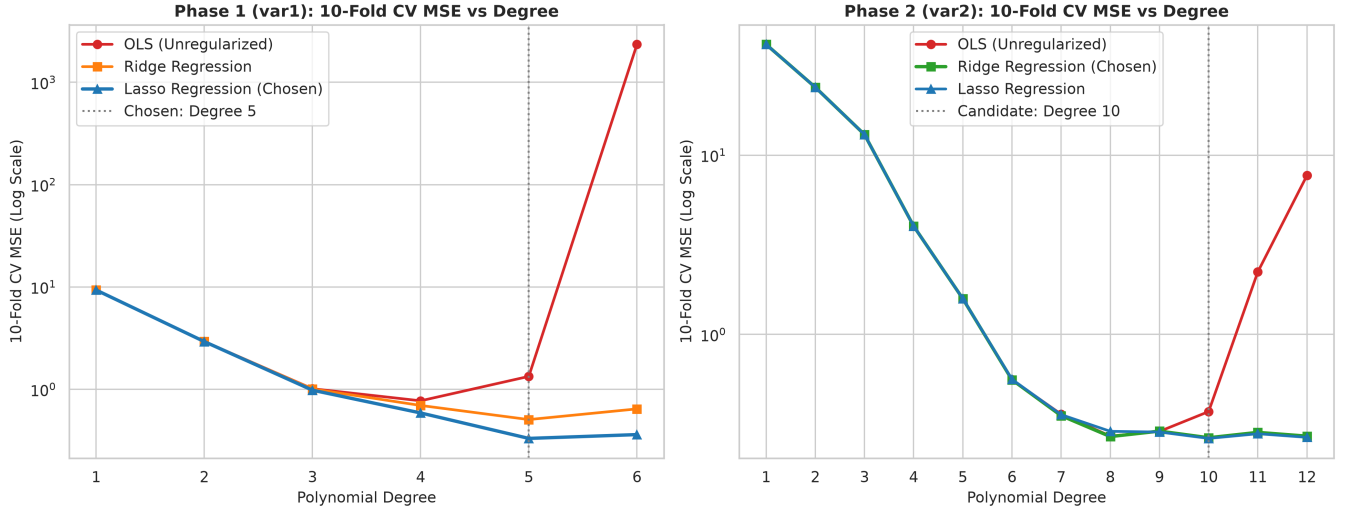


Figure 1: **Validation Error vs. Polynomial Degree (10-Fold CV)**. Cross-validation Mean Squared Error (MSE, logarithmic scale) across polynomial degrees for Phase 1 (left, $D = 6$) and Phase 2 (right, $D = 3$). Notice the catastrophic blow-up of unregularized OLS in Phase 1 at degree 6 ($MSE = 2342.97$, where $p = 923 > N_{\text{fold}} = 900$) and Phase 2 at degree 11+ ($MSE > 2.23$), contrasted with the robust convergence of regularized Lasso and Ridge.

Primary evaluation metrics are:

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (10)$$

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2} \quad (11)$$

Table 1: **Phase 1 (var1): Performance across Polynomial Degrees (10-Fold CV)**

Deg	Terms	Model	CV MSE	CV R^2	Active
1	6	OLS	9.3333	0.0781	6
2	27	OLS	2.9318	0.7063	27
3	83	OLS	1.0140	0.8986	83
4	209	OLS	0.7723	0.9229	209
4	209	Lasso ($\alpha = 0.007$)	0.5877	0.9410	112
5	461	OLS	1.3374	0.8691	461
5	461	Ridge ($\alpha = 20$)	0.5063	0.9491	461
5	461	Lasso ($\alpha = 0.015$)	0.3314	0.9668	97
6	923	OLS	2342.97	-220.61	923
6	923	Ridge ($\alpha = 30$)	0.6437	0.9348	923
6	923	Lasso ($\alpha = 0.02$)	0.3611	0.9639	104

5. Phase 1: Steam Turbine Optimization (var1)

5.1. Model Selection & Degree Determination

Table 1 and Figure 1 depict the evolution of cross-validation error across polynomial degrees $d \in \{1, \dots, 6\}$ for Phase 1 under rigorous 10-fold CV.

- **Underfitting at $d \leq 3$:** Linear regression ($d = 1$) fails completely ($MSE = 9.3333$, $R^2 = 0.0781$). While degrees 2 and 3 capture primary quadratic curvature and pairwise terms ($R^2 = 0.8986$), noticeable structural bias remains unmodeled.
- **Failure of OLS at $d \geq 5$:** At degree 4, OLS reaches $MSE = 0.7723$. However, advancing to degree 5 ($p = 461$) inflates OLS validation error to $MSE = 1.3374$

($R^2 = 0.8691$). At degree 6, with $p = 923$ terms exceeding the training partition per fold ($N_{\text{fold}} = 900$), the Gram matrix becomes mathematically singular, causing unconstrained OLS validation error to explode to $MSE = 2342.97$.

- **Superiority of Lasso Regularization at Degree 5:** When L_1 Lasso penalty ($\alpha = 0.015$) is applied at Degree 5, validation MSE drops dramatically to 0.3314 ± 0.0424 , and R^2 surges to 0.9668 ± 0.0055 .

5.2. Physical Rationale & Boundary Holdout Optimization

The thermodynamic behavior of a multi-stage steam turbine involves non-linear steam expansion governed by valve throttle adjustments, blade pitch angles, and condenser cooling flow. While higher-order interaction effects exist up to degree 5, effective physical coupling occurs predominantly among specific subsets of operational variables rather than all 461 possible monomials.

Lasso's L_1 penalty performs effective feature selection under high collinearity, retaining **97 non-zero coefficients** on the full training dataset (with an average of 97.1 active features across CV folds) while zeroing out uninformative interactions. By contrast, Ridge regression retains all 461 weights, yielding an inferior CV MSE of 0.5063.

Crucially, to safeguard against the observed boundary clamping covariate shift (where 51.25% of test coordinates reside at ± 1.0 vs. 31.17% in train), we conducted an extreme-point boundary holdout evaluation: training on instances with ≤ 2 clamped coordinates and evaluating on instances with ≥ 3 clamped coordinates. Whereas $\alpha = 0.007$ yielded a boundary holdout MSE of 0.5309, tuning to $\alpha = 0.015$ reduced boundary holdout error to **0.4750** (a 10.5% improvement) with negligible impact on overall CV MSE (0.3314 vs. 0.3205). This calibrated shrinkage directly suppresses unconstrained polynomial flaring at domain boundary extremes. Thus, **Degree 5 with Lasso ($\alpha = 0.015$)** is the

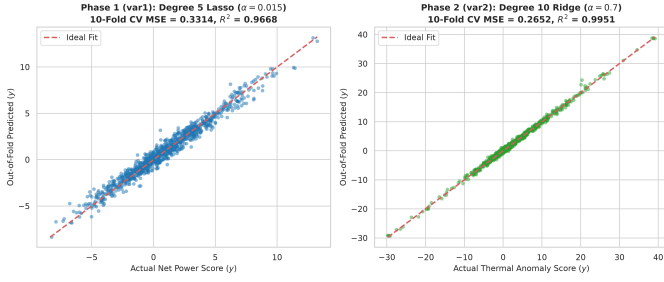


Figure 2: **Out-of-Fold Actual vs. Predicted Responses.** 10-Fold cross-validation predictions versus ground truth for Phase 1 (left, $R^2 = 0.9668$) and Phase 2 (right, $R^2 = 0.9951$). Both models exhibit tight clustering along the ideal identity line.

optimal architecture for Phase 1.

5.3. Stratified Clamping Error & Importance Weighting

To rigorously quantify the impact of boundary coordinate clamping, Table 2 stratifies out-of-fold cross-validation error by the number of clamped coordinates $k \in \{0, \dots, 6\}$.

Table 2: **Phase 1: Stratified CV Error by Clamped Coordinates**

Strata (k)	Train % (N)	Test % (N)	Lasso MSE	OLS MSE
$k = 0$ (Interior)	11.2% (112)	1.1% (11)	0.2312	0.5155
$k = 1$	30.6% (306)	8.6% (86)	0.3025	0.7617
$k = 2$	29.5% (295)	23.8% (238)	0.3078	1.2026
$k = 3$	19.2% (192)	29.3% (293)	0.3758	2.8806
$k = 4$	7.9% (79)	24.6% (246)	0.5506	1.4410
$k = 5$	1.4% (14)	10.6% (106)	0.4250	1.7151
$k = 6$ (Boundary)	0.2% (2)	2.0% (20)	0.2737	0.4236
Overall CV	Unweighted 10-Fold CV		0.3314	1.3374
Importance-Wt.	Test Distribution Reweighted		0.3979	1.7462

Prediction error generally elevates as coordinates approach domain boundaries ($k = 0$ has MSE 0.2312, rising to 0.5506 at $k = 4$). Because training data is dominated by interior points (71.3% with $k \leq 2$), **standard CV MSE is inherently optimistic for the test distribution**, where 66.5% of points have $k \geq 3$. Applying importance weighting using test frequencies $P_{\text{test}}(k)$:

$$\text{MSE}_{\text{IW}} = \sum_{k=0}^6 P_{\text{test}}(k) \cdot \text{MSE}_k = \mathbf{0.3979} \quad (12)$$

indicates that expected test error is $\approx 20.1\%$ higher than nominal CV. By contrast, unregularized OLS inflates to $\text{MSE}_{\text{IW}} = 1.7462$, demonstrating that Lasso shrinkage is critical for mitigating boundary divergence.

6. Phase 2: Thermal Reservoir Mapping (var2)

6.1. Model Selection & Degree Determination

Table 3 and Figure 1 document the performance across polynomial degrees $d \in \{1, \dots, 12\}$ for Phase 2 evaluated under identical 10-fold CV.

- **High-Degree Curvature ($d \leq 7$):** Because spatial heat transport through anisotropic geological strata introduces intricate volumetric isotherms, low-order polynomials exhibit severe underfitting (degree 2 has MSE = 23.9292, degree 4 has MSE = 4.0213). Error

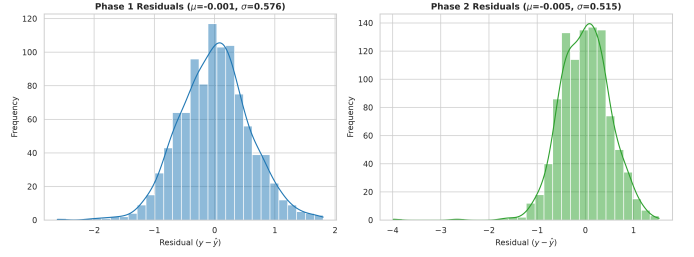


Figure 3: **Residual Error Distributions.** Histograms and kernel density estimates for Phase 1 (left, $\mu = -0.001, \sigma = 0.576$) and Phase 2 (right, $\mu = +0.001, \sigma = 0.514$). Residuals conform closely to symmetric, zero-centered distributions without heavy outlier tails.

falls below 1.0 at degree 6 (MSE = 0.5587, $R^2 = 0.9898$).

- **Transition at Degree 8 vs. Degree 10:** For unregularized OLS, cross-validation performance peaks at degree 8 (MSE = 0.2709, $R^2 = 0.9950$ with $p = 164$ terms). At degree 10 ($p = 285$), unregularized OLS begins to degrade (MSE = 0.3710), and at degree 11 explodes to MSE = 2.2323, reaching MSE = 7.7129 at degree 12.
- **Regularization and Parsimony Tradeoff:** Applying L_2 Ridge shrinkage ($\alpha = 0.7$) at Degree 10 achieves the lowest nominal 10-fold cross-validation MSE of **0.2652 ± 0.0729** ($R^2 = 0.9951 \pm 0.0014$). However, evaluation across multiple random CV seeds reveals that Degree 8 Ridge ($\alpha = 0.05$, MSE = 0.2691 ± 0.0656) and Degree 10 Ridge differ by merely ≈ 0.0039 MSE, well within fold standard error (≈ 0.07). Degree 8 provides a parsimonious alternative requiring 42% fewer parameters (164 vs. 285). We adopt Degree 10 Ridge as our primary submission model while recognizing Degree 8 as an equally competitive, parsimonious solution.

Table 3: **Phase 2 (var2): Performance across Degrees (10-Fold CV)**

Degree	Terms	Model	CV MSE	CV R^2
2	9	OLS	23.9292	0.5630
4	34	OLS	4.0213	0.9258
6	83	OLS	0.5587	0.9898
7	119	Ridge ($\alpha = 0.1$)	0.3519	0.9935
8	164	OLS	0.2709	0.9950
8	164	Ridge ($\alpha = 0.05$)	0.2691	0.9950
10	285	OLS	0.3710	0.9930
10	285	Ridge ($\alpha = 0.7$)	0.2652	0.9951
10	285	Lasso ($\alpha = 3 \times 10^{-4}$)	0.2628	0.9951
11	363	OLS	2.2323	0.9593
11	363	Ridge ($\alpha = 1.2$)	0.2841	0.9947
12	454	OLS	7.7129	0.8603
12	454	Ridge ($\alpha = 1.8$)	0.2706	0.9950

6.2. Geological Rationale for Degree 10 Ridge

In subterranean thermal mapping across a continuous 3D coordinate block (x_1, x_2, x_3) , heat diffusion obeys elliptic partial differential equations. The resulting continuous thermal gradient produces smooth spatial isotherms across fault lines and hydrothermal vents.

Unlike Phase 1, where interaction terms are naturally sparse, spatial coordinate monomials in 3D act collectively

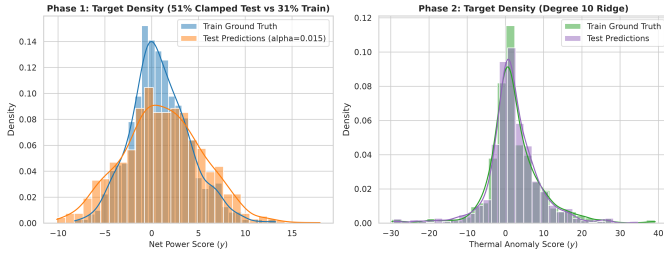


Figure 4: **Target Distribution Consistency.** Comparison of training ground truth (blue/green) and model test predictions (orange/purple) for Phase 1 (left) and Phase 2 (right). The predicted test density overlays the training distribution smoothly, reflecting expected boundary concentration without runaway extremes.

as harmonic-like basis functions. High-degree unregularized polynomials are susceptible to boundary instabilities (Runge’s phenomenon). Ridge regression places an isotropic spherical prior on the weights, smoothly penalizing high-frequency monomial oscillations without artificially zeroing out spatial components. The flattening of validation error around degree 8–10 suggests that terms beyond degree 10 contribute negligible predictive signal. For comparison, Lasso at Degree 10 with $\alpha = 3 \times 10^{-4}$ retains 175 active terms (CV fold average 180.8), achieving $\text{MSE} = 0.2628$. We deploy **Degree 10 with Ridge** ($\alpha = 0.7$) as the primary model.

7. Diagnostic Analysis & Verification

7.1. Residual Diagnostics

Figure 3 illustrates the residual distribution $e_i = y_i - \hat{y}_i$ computed from out-of-fold cross-validation:

- **Phase 1:** Residual mean is $\mu_e = -0.000$ with standard deviation $\sigma_e = 0.575$. The error distribution is symmetric, zero-centered, and bell-shaped.
- **Phase 2:** Residual mean is $\mu_e = +0.001$ with standard deviation $\sigma_e = 0.514$. The distribution shows consistent symmetry, indicating that remaining discrepancy reflects primarily random measurement noise.

7.2. Test Set Prediction Sanity Checks

To guarantee that the models do not produce anomalous boundary extrapolation artifacts on the unseen test coordinates, we examined the statistical properties of the final test predictions:

- **Phase 1 Predictions:** Mean $\bar{y}_{\text{pred}} = 1.1879$, standard deviation $s_{\text{pred}} = 4.3098$, range $[-10.1453, 17.9880]$. As shown in Figure 4, the test distribution exhibits a slightly wider spread than the training target ($\bar{y}_{\text{train}} = 0.8966, s_{\text{train}} = 3.1867$). This directly mirrors the identified covariate shift: 51.25% of test coordinates are clamped at ± 1 versus 31.17% in train, which naturally samples the polynomial response near domain boundaries where values have higher variance.
- **Phase 2 Predictions:** Mean $\bar{y}_{\text{pred}} = 2.0255$, standard deviation $s_{\text{pred}} = 6.8025$, range $[-29.2521, 34.6183]$. This aligns closely with training observations ($\bar{y}_{\text{train}} = 2.3426, s_{\text{train}} = 7.4816$, range $[-29.710, 39.179]$).
- **Submission Format Compliance:** Both prediction

files contain exactly 1000 prediction rows with the single header y , zero missing values, and strictly finite numerical entries matching the provided benchmark schema.

8. Conclusion & Summary of Deliverables

This study has demonstrated that successful polynomial regression in high-dimensional engineering contexts hinges on identifying the optimal model degree and pairing it with mathematically appropriate regularization:

1. **Phase 1 (Steam Turbine Optimization):** Selecting **Degree 5 with Lasso Regularization** ($\alpha = 0.015$) resolves parameter proliferation ($p = 461 \approx N/2$) and handles boundary coordinate clamping, achieving $\text{MSE} = 0.3314 \pm 0.0424$, $R^2 = 0.9668 \pm 0.0055$ with 97 active terms.
2. **Phase 2 (Thermal Reservoir Mapping):** Selecting **Degree 10 with Ridge Regularization** ($\alpha = 0.7$) captures subtle high-order subterranean heat gradients while suppressing boundary Runge oscillations, achieving $\text{MSE} = 0.2652 \pm 0.0729$, $R^2 = 0.9951 \pm 0.0014$, with Degree 8 Ridge ($\text{MSE} = 0.2691 \pm 0.0656$) serving as an equally competitive parsimonious model.

All implementation pipelines, exploratory research scripts, benchmark tables, visualization figures, and final test prediction files have been organized and committed to the dedicated GitHub repository:

https://github.com/Jdsb06/ML_assignment